

BOA VISTA, Brazil

WMO: 820220

Lat: 2.846N

Lon: 60.690W

Elev: 84

StdP: 100.32

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	22.8	15.0	10.8	32.5	16.0	11.5	32.3	8.4	27.0	7.1	28.4	2.3	60	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	9.3	36.0	24.5	35.2	24.3	34.9	24.2	26.2	31.9	25.9	31.6	25.7	31.1	4.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.4	27.4	24.7	20.0	27.3	24.2	19.3	27.0	81.5	32.0	80.3	32.0	79.2	31.3	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.4	7.5	20.9	37.1	1.1	1.0	20.1	37.8	19.5	38.5	18.9	39.0	18.1	39.8		
(5)			18.6	27.7	1.7	1.2	17.4	28.6	16.4	29.3	15.4	29.9	14.2	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	28.7	28.8	29.3	28.9	27.7	27.4	27.3	28.2	29.2	29.5	29.3	28.8	
	DBStd	1.52	1.04	1.08	1.24	1.52	1.56	1.26	1.26	1.42	1.34	1.45	1.40	1.21	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6782	579	527	598	566	549	521	536	563	574	604	579	583	
	CDD18.3	3740	321	294	340	316	291	271	277	305	324	346	329	325	
	CDH23.3	42456	3775	3516	4263	3709	2922	2529	2507	3166	3870	4314	4063	3822	
	CDH26.7	18704	1692	1603	2063	1675	1064	836	818	1277	1830	2147	1974	1726	
Wind	WSAvg	3.3	4.7	4.9	4.8	3.7	2.8	2.3	2.1	2.2	2.5	2.9	3.1	3.9	
Precipitation	PrecAvg	1547	22	19	51	142	265	318	293	197	98	58	67	44	
	PrecMax	2213	99	76	297	457	565	650	556	371	251	212	269	234	
	PrecMin	228	0	0	0	0	19	24	17	9	0	0	1	0	
	PrecStd	457	25	19	72	113	133	145	121	87	70	51	58	49	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	35.8	36.2	35.9	34.5	33.8	33.7	35.1	36.1	36.2	36.2	35.8	
		MCWB	24.1	24.2	23.6	23.7	24.4	25.2	24.8	25.0	25.2	24.7	25.2	24.7	
	2%	DB	34.2	34.2	35.1	34.9	33.6	32.8	32.5	34.1	35.1	35.7	35.2	34.7	
		MCWB	23.5	23.2	23.5	23.7	24.3	24.8	24.6	24.9	24.7	24.6	24.7	24.5	
	5%	DB	33.5	33.8	34.2	34.0	32.5	31.9	31.8	33.1	34.2	35.0	34.6	33.8	
		MCWB	23.4	23.1	23.2	23.6	24.5	24.7	24.5	24.7	24.4	24.4	24.4	24.1	
	10%	DB	32.9	33.0	33.7	33.1	31.4	30.9	30.8	32.1	33.8	34.1	33.9	32.9	
		MCWB	23.3	23.1	23.2	23.8	24.5	24.6	24.5	24.6	24.4	24.2	24.1	23.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.6	25.7	25.8	26.2	26.4	26.2	25.9	26.3	26.2	26.2	26.3	26.2
			MCDB	31.9	31.7	31.9	30.5	30.8	30.3	30.6	32.0	32.6	33.3	33.2	32.4
2%		WB	25.0	25.0	25.1	25.6	25.9	25.7	25.5	25.8	25.8	25.7	25.7	25.6	
		MCDB	31.3	31.3	31.0	30.2	30.1	29.9	29.9	31.2	32.1	32.4	32.5	31.5	
5%		WB	24.6	24.5	24.6	25.2	25.5	25.3	25.1	25.5	25.5	25.4	25.3	25.1	
		MCDB	30.6	30.7	30.6	29.8	29.5	29.3	29.3	30.5	31.5	31.8	31.6	30.8	
10%		WB	24.2	24.1	24.2	24.8	25.1	25.0	24.8	25.1	25.2	25.1	25.0	24.7	
		MCDB	30.0	30.1	30.2	29.5	29.0	28.9	28.7	29.8	30.7	31.1	30.8	30.2	
Mean Daily Temperature Range		5% DB	MDBR	8.7	8.7	8.8	8.2	7.1	7.1	7.3	8.2	9.1	9.3	9.0	8.6
			MCDBR	9.5	9.6	9.6	9.3	8.6	8.3	8.4	9.3	9.7	9.9	9.8	9.5
	MCWBR		2.3	2.4	2.3	2.2	2.3	2.4	2.3	2.4	2.4	2.2	2.2	2.2	
	5% WB	MCDBR	8.4	8.4	8.5	8.0	7.4	7.4	7.6	8.5	9.2	9.4	9.2	8.5	
		MCWBR	2.5	2.7	2.6	2.5	2.5	2.5	2.5	2.6	2.5	2.5	2.4	2.4	
Clear-Sky Solar Irradiance	taub	0.390	0.404	0.409	0.399	0.393	0.397	0.407	0.422	0.433	0.422	0.418	0.407		
	taud	2.516	2.477	2.469	2.512	2.527	2.526	2.488	2.422	2.373	2.414	2.425	2.464		
	Ebn at Noon	934	925	915	906	890	874	870	873	880	897	900	912		
	Edh at Noon	109	116	117	109	104	102	107	118	126	121	119	114		
All-Sky Solar Radiation	RadAvg	5.23	5.21	5.34	5.00	4.46	4.64	4.76	5.34	5.90	5.76	5.37	4.98		
	RadStd	0.28	0.30	0.30	0.34	0.40	0.32	0.33	0.34	0.27	0.23	0.29	0.26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		-0.42	-0.33	-1.45	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page